

4. Bud rate

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1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

2 Data formatting

Create a new data frame adapted for generalized linear mixed-effects models (GLMM).

```

buds_df <-

  #' Load the raw data frame.
  df %>%

  #' Reshape the weekly bud counts from wide to long format.
  pivot_longer(cols = starts_with("buds_"),
               names_to = c(".value", "week"),
               names_sep = "_",) %>%

  #' Convert week and bud count to integers.
  mutate(week = as.integer(week),
         buds = as.integer(buds),
  #' Create a centered variable for the week.
         week_centered = week - mean(week, na.rm = TRUE)) %>%

  #' Remove weeks with missing bud count.
  filter(!is.na(buds)) %>%

  #' Keep only the variables needed for the analysis.
  select(treatment_condition, lineage, tum_state, week, week_centered,
         buds, id_unique, parent_unique, replicate_unique)

```

Set reference levels for the categorical variables.

```

buds_df$lineage <- relevel(factor(buds_df$lineage), ref = "HO_MT")
buds_df$treatment_condition <- relevel(factor(buds_df$treatment_condition), ref =
"Control")
buds_df$tum_state <- relevel(factor(buds_df$tum_state), ref = "Healthy")

```

3 Model selection

3.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```

buds_pois_R_00 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition, REML =
TRUE, family = poisson, data = buds_df)
buds_pois_R_01 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
id_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_02 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition +
(week_centered | id_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_03 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_04 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_05 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
id_unique) + (1 | parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_06 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
id_unique) + (1 | replicate_unique), REML = TRUE, family = poisson, data = buds_df)

```

```
buds_pois_R_07 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 | id_unique) + (1 | parent_unique) + (1 | replicate_unique), REML = TRUE, family = poisson, data = buds_df)
```

```
AIC(buds_pois_R_00, buds_pois_R_01, buds_pois_R_02, buds_pois_R_03, buds_pois_R_04, buds_pois_R_05, buds_pois_R_06, buds_pois_R_07) %>% arrange(AIC)
```

```
##           df      AIC
## buds_pois_R_04 25 12602.13
## buds_pois_R_06 26 12604.13
## buds_pois_R_07 27 12606.13
## buds_pois_R_02 27 12616.80
## buds_pois_R_03 25 12625.02
## buds_pois_R_01 25 12625.68
## buds_pois_R_05 26 12626.02
## buds_pois_R_00 24 12628.14
```

The model with the replicate id as a random effect (buds_pois_R_04) has the lowest AIC, this random effect structure is therefore selected.

3.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```
buds_pois_F_00 <- glmmTMB(buds ~ 1 + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_01 <- glmmTMB(buds ~ week_centered + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_02 <- glmmTMB(buds ~ lineage + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_03 <- glmmTMB(buds ~ treatment_condition + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_04 <- glmmTMB(buds ~ tum_state + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_05 <- glmmTMB(buds ~ week_centered + lineage + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_06 <- glmmTMB(buds ~ week_centered + treatment_condition + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_07 <- glmmTMB(buds ~ week_centered + tum_state + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_08 <- glmmTMB(buds ~ lineage + treatment_condition + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_09 <- glmmTMB(buds ~ lineage + tum_state + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_10 <- glmmTMB(buds ~ treatment_condition + tum_state + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_11 <- glmmTMB(buds ~ week_centered * lineage + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_12 <- glmmTMB(buds ~ week_centered * treatment_condition + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_13 <- glmmTMB(buds ~ week_centered * tum_state + (1 | replicate_unique), REML = FALSE, family = poisson, data = buds_df)
```

```

buds_pois_F_14 <- glmmTMB(buds ~ lineage * treatment_condition + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_15 <- glmmTMB(buds ~ lineage * tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_16 <- glmmTMB(buds ~ treatment_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_17 <- glmmTMB(buds ~ week_centered + lineage + treatment_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_18 <- glmmTMB(buds ~ week_centered + lineage + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_19 <- glmmTMB(buds ~ week_centered + treatment_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_20 <- glmmTMB(buds ~ lineage + treatment_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_21 <- glmmTMB(buds ~ week_centered * lineage + treatment_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_22 <- glmmTMB(buds ~ week_centered * treatment_condition + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_23 <- glmmTMB(buds ~ week_centered * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_24 <- glmmTMB(buds ~ week_centered * lineage + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_25 <- glmmTMB(buds ~ week_centered * treatment_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_26 <- glmmTMB(buds ~ week_centered * tum_state + treatment_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_27 <- glmmTMB(buds ~ lineage * treatment_condition + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_28 <- glmmTMB(buds ~ lineage * tum_state + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_29 <- glmmTMB(buds ~ treatment_condition * tum_state + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_30 <- glmmTMB(buds ~ lineage * treatment_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_31 <- glmmTMB(buds ~ lineage * tum_state + treatment_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_32 <- glmmTMB(buds ~ treatment_condition * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_33 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_34 <- glmmTMB(buds ~ week_centered * lineage * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_35 <- glmmTMB(buds ~ week_centered * treatment_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_36 <- glmmTMB(buds ~ lineage * treatment_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)

```

```

AIC(buds_pois_F_00, buds_pois_F_01, buds_pois_F_02, buds_pois_F_03, buds_pois_F_04,
buds_pois_F_05, buds_pois_F_06, buds_pois_F_07, buds_pois_F_08, buds_pois_F_09,
buds_pois_F_10, buds_pois_F_11, buds_pois_F_12, buds_pois_F_13, buds_pois_F_14,
buds_pois_F_15, buds_pois_F_16, buds_pois_F_17, buds_pois_F_18, buds_pois_F_19,
buds_pois_F_20, buds_pois_F_21, buds_pois_F_22, buds_pois_F_23, buds_pois_F_24,
buds_pois_F_25, buds_pois_F_26, buds_pois_F_27, buds_pois_F_28, buds_pois_F_29,
buds_pois_F_30, buds_pois_F_31, buds_pois_F_32, buds_pois_F_33, buds_pois_F_34,

```

```
buds_pois_F_35, buds_pois_F_36) %>% arrange(AIC)
```

```
##           df      AIC
## buds_pois_F_33 25 12489.87
## buds_pois_F_21 14 12494.81
## buds_pois_F_11 13 12500.31
## buds_pois_F_24 14 12500.54
## buds_pois_F_34 19 12509.36
## buds_pois_F_27 14 12561.96
## buds_pois_F_22 10 12563.02
## buds_pois_F_17  9 12564.93
## buds_pois_F_05  8 12570.68
## buds_pois_F_18  9 12570.97
## buds_pois_F_23 10 12572.75
## buds_pois_F_28 11 12574.36
## buds_pois_F_14 13 12862.38
## buds_pois_F_30 14 12862.49
## buds_pois_F_08  8 12866.84
## buds_pois_F_32 10 12867.85
## buds_pois_F_20  9 12868.47
## buds_pois_F_31 11 12870.53
## buds_pois_F_36 19 12871.30
## buds_pois_F_02  7 12874.06
## buds_pois_F_09  8 12874.72
## buds_pois_F_15 10 12878.25
## buds_pois_F_25  6 12881.20
## buds_pois_F_29  6 12882.98
## buds_pois_F_19  5 12883.40
## buds_pois_F_35  9 12883.43
## buds_pois_F_26  6 12885.35
## buds_pois_F_07  4 12893.10
## buds_pois_F_12  5 12894.87
## buds_pois_F_13  5 12895.04
## buds_pois_F_06  4 12896.94
## buds_pois_F_01  3 12902.29
## buds_pois_F_16  5 13173.44
## buds_pois_F_10  4 13174.39
## buds_pois_F_04  3 13186.02
## buds_pois_F_03  3 13189.04
## buds_pois_F_00  2 13195.82
```

The model with the interaction between the week, the lineage and the treatment condition has the lowest AIC and is therefore selected (buds_pois_F_33).

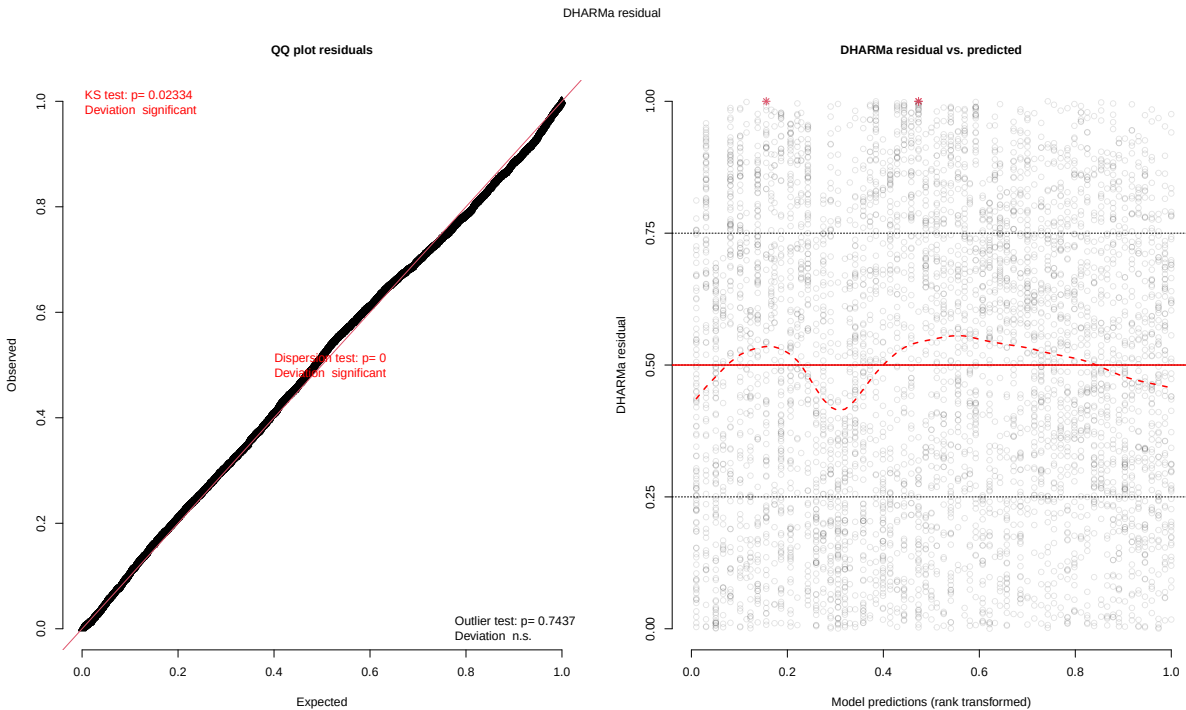
3.3 Goodness-of-fit evaluation

Refit the selected model.

```
buds_pois_mod_33 <- glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
```

Assess goodness-of-fit of the selected model using simulated residuals.

```
buds_pois_mod_33_res <- simulateResiduals(buds_pois_mod_33, plot = TRUE)
```



3.4 Selected model summary

Releveling of "lineage" to obtain all pairwise comparisons from the same model.

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_MT")
summary(glmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * treatment_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1    4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
```

##	Estimate
## (Intercept)	0.457269
## week_centered	0.040455
## lineageHC_MT	-0.573663
## lineageHO_SPC	-0.256052
## lineageHO_SPT	-0.092013
## lineageHO_VLN	-1.013640
## lineageHV_GAL	-0.061518
## treatment_conditionShaking treatment	-0.063490
## week_centered:lineageHC_MT	0.227553
## week_centered:lineageHO_SPC	0.101016
## week_centered:lineageHO_SPT	0.126876
## week_centered:lineageHO_VLN	0.178343
## week_centered:lineageHV_GAL	0.140067
## week_centered:treatment_conditionShaking treatment	-0.007714
## lineageHC_MT:treatment_conditionShaking treatment	-0.209042
## lineageHO_SPC:treatment_conditionShaking treatment	0.140629
## lineageHO_SPT:treatment_conditionShaking treatment	0.030166
## lineageHO_VLN:treatment_conditionShaking treatment	-0.119088
## lineageHV_GAL:treatment_conditionShaking treatment	-0.061672
## week_centered:lineageHC_MT:treatment_conditionShaking treatment	-0.007118
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment	-0.065524
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment	0.042134
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment	0.034530
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment	-0.098026
##	Std. Error
## (Intercept)	0.044793
## week_centered	0.023019
## lineageHC_MT	0.086495
## lineageHO_SPC	0.066325
## lineageHO_SPT	0.066574
## lineageHO_VLN	0.074958
## lineageHV_GAL	0.064093
## treatment_conditionShaking treatment	0.057329
## week_centered:lineageHC_MT	0.044824
## week_centered:lineageHO_SPC	0.034484
## week_centered:lineageHO_SPT	0.034150
## week_centered:lineageHO_VLN	0.040251
## week_centered:lineageHV_GAL	0.033253
## week_centered:treatment_conditionShaking treatment	0.033359
## lineageHC_MT:treatment_conditionShaking treatment	0.125988
## lineageHO_SPC:treatment_conditionShaking treatment	0.083929
## lineageHO_SPT:treatment_conditionShaking treatment	0.084802
## lineageHO_VLN:treatment_conditionShaking treatment	0.103689
## lineageHV_GAL:treatment_conditionShaking treatment	0.084267
## week_centered:lineageHC_MT:treatment_conditionShaking treatment	0.069369
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment	0.048808
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment	0.048697
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment	0.059922
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment	0.049032
##	z value
## (Intercept)	10.209
## week_centered	1.757
## lineageHC_MT	-6.632

```

## lineageHO_SPC -3.861
## lineageHO_SPT -1.382
## lineageHO_VLN -13.523
## lineageHV_GAL -0.960
## treatment_conditionShaking treatment -1.107
## week_centered:lineageHC_MT 5.077
## week_centered:lineageHO_SPC 2.929
## week_centered:lineageHO_SPT 3.715
## week_centered:lineageHO_VLN 4.431
## week_centered:lineageHV_GAL 4.212
## week_centered:treatment_conditionShaking treatment -0.231
## lineageHC_MT:treatment_conditionShaking treatment -1.659
## lineageHO_SPC:treatment_conditionShaking treatment 1.676
## lineageHO_SPT:treatment_conditionShaking treatment 0.356
## lineageHO_VLN:treatment_conditionShaking treatment -1.149
## lineageHV_GAL:treatment_conditionShaking treatment -0.732
## week_centered:lineageHC_MT:treatment_conditionShaking treatment -0.103
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment -1.343
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.865
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.576
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -1.999
## Pr(>|z|)
## (Intercept) < 2e-16 ***
## week_centered 0.078845 .
## lineageHC_MT 3.30e-11 ***
## lineageHO_SPC 0.000113 ***
## lineageHO_SPT 0.166935
## lineageHO_VLN < 2e-16 ***
## lineageHV_GAL 0.337142
## treatment_conditionShaking treatment 0.268097
## week_centered:lineageHC_MT 3.84e-07 ***
## week_centered:lineageHO_SPC 0.003397 **
## week_centered:lineageHO_SPT 0.000203 ***
## week_centered:lineageHO_VLN 9.39e-06 ***
## week_centered:lineageHV_GAL 2.53e-05 ***
## week_centered:treatment_conditionShaking treatment 0.817126
## lineageHC_MT:treatment_conditionShaking treatment 0.097072 .
## lineageHO_SPC:treatment_conditionShaking treatment 0.093821 .
## lineageHO_SPT:treatment_conditionShaking treatment 0.722053
## lineageHO_VLN:treatment_conditionShaking treatment 0.250756
## lineageHV_GAL:treatment_conditionShaking treatment 0.464253
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.918272
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.179432
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.386910
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.564451
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.045582 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPC")
summary(glmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))

```

```
## Family: poisson ( log )
```

```

## Formula:
## buds ~ week_centered * lineage * treatment_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
##   Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##                                     Estimate
## (Intercept)                        0.20122
## week_centered                      0.14147
## lineageHO_MT                      0.25605
## lineageHC_MT                     -0.31761
## lineageHO_SPT                     0.16404
## lineageHO_VLN                    -0.75759
## lineageHV_GAL                     0.19453
## treatment_conditionShaking treatment 0.07714
## week_centered:lineageHO_MT        -0.10102
## week_centered:lineageHC_MT         0.12654
## week_centered:lineageHO_SPT        0.02586
## week_centered:lineageHO_VLN        0.07733
## week_centered:lineageHV_GAL        0.03905
## week_centered:treatment_conditionShaking treatment -0.07324
## lineageHO_MT:treatment_conditionShaking treatment -0.14063
## lineageHC_MT:treatment_conditionShaking treatment -0.34967
## lineageHO_SPT:treatment_conditionShaking treatment -0.11046
## lineageHO_VLN:treatment_conditionShaking treatment -0.25972
## lineageHV_GAL:treatment_conditionShaking treatment -0.20230
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.06552
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.05841
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.10766
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.10005
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -0.03250
##
##                                     Std. Error
## (Intercept)                        0.04895
## week_centered                      0.02568
## lineageHO_MT                      0.06633
## lineageHC_MT                      0.08872
## lineageHO_SPT                     0.06944
## lineageHO_VLN                     0.07751
## lineageHV_GAL                     0.06706
## treatment_conditionShaking treatment 0.06130
## week_centered:lineageHO_MT        0.03448
## week_centered:lineageHC_MT        0.04624
## week_centered:lineageHO_SPT       0.03599
## week_centered:lineageHO_VLN       0.04183
## week_centered:lineageHV_GAL       0.03514

```

```

## week_centered:treatment_conditionShaking treatment      0.03563
## lineageHO_MT:treatment_conditionShaking treatment      0.08393
## lineageHC_MT:treatment_conditionShaking treatment      0.12785
## lineageHO_SPT:treatment_conditionShaking treatment      0.08753
## lineageHO_VLN:treatment_conditionShaking treatment      0.10594
## lineageHV_GAL:treatment_conditionShaking treatment      0.08701
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.04881
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.07049
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.05028
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.06121
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.05060
##
## z value
## (Intercept)      4.111
## week_centered      5.510
## lineageHO_MT      3.861
## lineageHC_MT     -3.580
## lineageHO_SPT      2.362
## lineageHO_VLN     -9.774
## lineageHV_GAL      2.901
## treatment_conditionShaking treatment      1.258
## week_centered:lineageHO_MT     -2.929
## week_centered:lineageHC_MT      2.736
## week_centered:lineageHO_SPT      0.718
## week_centered:lineageHO_VLN      1.849
## week_centered:lineageHV_GAL      1.111
## week_centered:treatment_conditionShaking treatment     -2.056
## lineageHO_MT:treatment_conditionShaking treatment     -1.676
## lineageHC_MT:treatment_conditionShaking treatment     -2.735
## lineageHO_SPT:treatment_conditionShaking treatment     -1.262
## lineageHO_VLN:treatment_conditionShaking treatment     -2.452
## lineageHV_GAL:treatment_conditionShaking treatment     -2.325
## week_centered:lineageHO_MT:treatment_conditionShaking treatment  1.343
## week_centered:lineageHC_MT:treatment_conditionShaking treatment  0.829
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment  2.141
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment  1.634
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -0.642
##
## Pr(>|z|)
## (Intercept)      3.94e-05 ***
## week_centered      3.59e-08 ***
## lineageHO_MT      0.000113 ***
## lineageHC_MT      0.000344 ***
## lineageHO_SPT      0.018155 *
## lineageHO_VLN      < 2e-16 ***
## lineageHV_GAL      0.003721 **
## treatment_conditionShaking treatment      0.208241
## week_centered:lineageHO_MT      0.003397 **
## week_centered:lineageHC_MT      0.006213 **
## week_centered:lineageHO_SPT      0.472475
## week_centered:lineageHO_VLN      0.064496 .
## week_centered:lineageHV_GAL      0.266492
## week_centered:treatment_conditionShaking treatment      0.039815 *
## lineageHO_MT:treatment_conditionShaking treatment      0.093821 .
## lineageHC_MT:treatment_conditionShaking treatment      0.006238 **
## lineageHO_SPT:treatment_conditionShaking treatment      0.206966

```

```
## lineageHO_VLN:treatment_conditionShaking treatment 0.014220 *
## lineageHV_GAL:treatment_conditionShaking treatment 0.020070 *
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.179432
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.407332
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.032252 *
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.102159
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.520692
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPT")
summary(glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * treatment_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1    4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##                                     Estimate
## (Intercept)                        0.365256
## week_centered                      0.167331
## lineageHO_SPC                     -0.164039
## lineageHO_MT                       0.092013
## lineageHC_MT                     -0.481650
## lineageHO_VLN                    -0.921627
## lineageHV_GAL                     0.030494
## treatment_conditionShaking treatment -0.033324
## week_centered:lineageHO_SPC        -0.025860
## week_centered:lineageHO_MT        -0.126876
## week_centered:lineageHC_MT         0.100677
## week_centered:lineageHO_VLN        0.051467
## week_centered:lineageHV_GAL        0.013190
## week_centered:treatment_conditionShaking treatment 0.034420
## lineageHO_SPC:treatment_conditionShaking treatment 0.110464
## lineageHO_MT:treatment_conditionShaking treatment -0.030166
## lineageHC_MT:treatment_conditionShaking treatment -0.239207
## lineageHO_VLN:treatment_conditionShaking treatment -0.149254
## lineageHV_GAL:treatment_conditionShaking treatment -0.091837
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment -0.107659
## week_centered:lineageHO_MT:treatment_conditionShaking treatment -0.042134
## week_centered:lineageHC_MT:treatment_conditionShaking treatment -0.049252
```

```

## week_centered:lineageHO_VLN:treatment_conditionShaking treatment -0.007605
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -0.140161
## Std. Error
## (Intercept) 0.049287
## week_centered 0.025225
## lineageHO_SPC 0.069436
## lineageHO_MT 0.066574
## lineageHC_MT 0.088903
## lineageHO_VLN 0.077724
## lineageHV_GAL 0.067302
## treatment_conditionShaking treatment 0.062486
## week_centered:lineageHO_SPC 0.035994
## week_centered:lineageHO_MT 0.034150
## week_centered:lineageHC_MT 0.045995
## week_centered:lineageHO_VLN 0.041552
## week_centered:lineageHV_GAL 0.034816
## week_centered:treatment_conditionShaking treatment 0.035476
## lineageHO_SPC:treatment_conditionShaking treatment 0.087534
## lineageHO_MT:treatment_conditionShaking treatment 0.084802
## lineageHC_MT:treatment_conditionShaking treatment 0.128442
## lineageHO_VLN:treatment_conditionShaking treatment 0.106626
## lineageHV_GAL:treatment_conditionShaking treatment 0.087829
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.050278
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.048697
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.070412
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.061126
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.050497
## z value
## (Intercept) 7.411
## week_centered 6.633
## lineageHO_SPC -2.362
## lineageHO_MT 1.382
## lineageHC_MT -5.418
## lineageHO_VLN -11.858
## lineageHV_GAL 0.453
## treatment_conditionShaking treatment -0.533
## week_centered:lineageHO_SPC -0.718
## week_centered:lineageHO_MT -3.715
## week_centered:lineageHC_MT 2.189
## week_centered:lineageHO_VLN 1.239
## week_centered:lineageHV_GAL 0.379
## week_centered:treatment_conditionShaking treatment 0.970
## lineageHO_SPC:treatment_conditionShaking treatment 1.262
## lineageHO_MT:treatment_conditionShaking treatment -0.356
## lineageHC_MT:treatment_conditionShaking treatment -1.862
## lineageHO_VLN:treatment_conditionShaking treatment -1.400
## lineageHV_GAL:treatment_conditionShaking treatment -1.046
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment -2.141
## week_centered:lineageHO_MT:treatment_conditionShaking treatment -0.865
## week_centered:lineageHC_MT:treatment_conditionShaking treatment -0.699
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment -0.124
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -2.776
## Pr(>|z|)
## (Intercept) 1.26e-13 ***

```

```
## week_centered 3.28e-11 ***
## lineageHO_SPC 0.018155 *
## lineageHO_MT 0.166935
## lineageHC_MT 6.04e-08 ***
## lineageHO_VLN < 2e-16 ***
## lineageHV_GAL 0.650481
## treatment_conditionShaking treatment 0.593822
## week_centered:lineageHO_SPC 0.472475
## week_centered:lineageHO_MT 0.000203 ***
## week_centered:lineageHC_MT 0.028607 *
## week_centered:lineageHO_VLN 0.215487
## week_centered:lineageHV_GAL 0.704790
## week_centered:treatment_conditionShaking treatment 0.331927
## lineageHO_SPC:treatment_conditionShaking treatment 0.206966
## lineageHO_MT:treatment_conditionShaking treatment 0.722053
## lineageHC_MT:treatment_conditionShaking treatment 0.062549 .
## lineageHO_VLN:treatment_conditionShaking treatment 0.161578
## lineageHV_GAL:treatment_conditionShaking treatment 0.295727
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.032252 *
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.386910
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.484246
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.900992
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.005510 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_VLN")
summary(glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * treatment_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1      4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept) -0.556370
## week_centered 0.218798
## lineageHO_SPT 0.921627
## lineageHO_SPC 0.757588
## lineageHO_MT 1.013640
## lineageHC_MT 0.439977
```

```

## lineageHV_GAL 0.952121
## treatment_conditionShaking treatment -0.182578
## week_centered:lineageHO_SPT -0.051467
## week_centered:lineageHO_SPC -0.077327
## week_centered:lineageHO_MT -0.178343
## week_centered:lineageHC_MT 0.049210
## week_centered:lineageHV_GAL -0.038276
## week_centered:treatment_conditionShaking treatment 0.026816
## lineageHO_SPT:treatment_conditionShaking treatment 0.149254
## lineageHO_SPC:treatment_conditionShaking treatment 0.259718
## lineageHO_MT:treatment_conditionShaking treatment 0.119088
## lineageHC_MT:treatment_conditionShaking treatment -0.089953
## lineageHV_GAL:treatment_conditionShaking treatment 0.057417
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.007605
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment -0.100054
## week_centered:lineageHO_MT:treatment_conditionShaking treatment -0.034530
## week_centered:lineageHC_MT:treatment_conditionShaking treatment -0.041648
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -0.132556
## Std. Error
## (Intercept) 0.060131
## week_centered 0.033019
## lineageHO_SPT 0.077724
## lineageHO_SPC 0.077511
## lineageHO_MT 0.074958
## lineageHC_MT 0.095343
## lineageHV_GAL 0.075608
## treatment_conditionShaking treatment 0.086399
## week_centered:lineageHO_SPT 0.041552
## week_centered:lineageHO_SPC 0.041827
## week_centered:lineageHO_MT 0.040251
## week_centered:lineageHC_MT 0.050688
## week_centered:lineageHV_GAL 0.040814
## week_centered:treatment_conditionShaking treatment 0.049778
## lineageHO_SPT:treatment_conditionShaking treatment 0.106626
## lineageHO_SPC:treatment_conditionShaking treatment 0.105936
## lineageHO_MT:treatment_conditionShaking treatment 0.103689
## lineageHC_MT:treatment_conditionShaking treatment 0.141613
## lineageHV_GAL:treatment_conditionShaking treatment 0.106192
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.061126
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.061215
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.059922
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.078592
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.061383
## z value
## (Intercept) -9.253
## week_centered 6.626
## lineageHO_SPT 11.858
## lineageHO_SPC 9.774
## lineageHO_MT 13.523
## lineageHC_MT 4.615
## lineageHV_GAL 12.593
## treatment_conditionShaking treatment -2.113
## week_centered:lineageHO_SPT -1.239
## week_centered:lineageHO_SPC -1.849

```

```

## week_centered:lineageHO_MT -4.431
## week_centered:lineageHC_MT 0.971
## week_centered:lineageHV_GAL -0.938
## week_centered:treatment_conditionShaking treatment 0.539
## lineageHO_SPT:treatment_conditionShaking treatment 1.400
## lineageHO_SPC:treatment_conditionShaking treatment 2.452
## lineageHO_MT:treatment_conditionShaking treatment 1.149
## lineageHC_MT:treatment_conditionShaking treatment -0.635
## lineageHV_GAL:treatment_conditionShaking treatment 0.541
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.124
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment -1.634
## week_centered:lineageHO_MT:treatment_conditionShaking treatment -0.576
## week_centered:lineageHC_MT:treatment_conditionShaking treatment -0.530
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -2.159
## Pr(>|z|)
## (Intercept) < 2e-16 ***
## week_centered 3.44e-11 ***
## lineageHO_SPT < 2e-16 ***
## lineageHO_SPC < 2e-16 ***
## lineageHO_MT < 2e-16 ***
## lineageHC_MT 3.94e-06 ***
## lineageHV_GAL < 2e-16 ***
## treatment_conditionShaking treatment 0.0346 *
## week_centered:lineageHO_SPT 0.2155
## week_centered:lineageHO_SPC 0.0645 .
## week_centered:lineageHO_MT 9.39e-06 ***
## week_centered:lineageHC_MT 0.3316
## week_centered:lineageHV_GAL 0.3483
## week_centered:treatment_conditionShaking treatment 0.5901
## lineageHO_SPT:treatment_conditionShaking treatment 0.1616
## lineageHO_SPC:treatment_conditionShaking treatment 0.0142 *
## lineageHO_MT:treatment_conditionShaking treatment 0.2508
## lineageHC_MT:treatment_conditionShaking treatment 0.5253
## lineageHV_GAL:treatment_conditionShaking treatment 0.5887
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.9010
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.1022
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.5645
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.5962
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.0308 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HC_MT")
summary(glmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))

```

```

## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * treatment_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728

```

```

##
## Random effects:
##
## Conditional model:
##   Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##                                     Estimate
## (Intercept)                       -0.116393
## week_centered                      0.268008
## lineageHO_VLN                     -0.439977
## lineageHO_SPT                      0.481650
## lineageHO_SPC                      0.317611
## lineageHO_MT                      0.573663
## lineageHV_GAL                     0.512144
## treatment_conditionShaking treatment -0.272531
## week_centered:lineageHO_VLN        -0.049210
## week_centered:lineageHO_SPT        -0.100677
## week_centered:lineageHO_SPC        -0.126537
## week_centered:lineageHO_MT        -0.227553
## week_centered:lineageHV_GAL        -0.087487
## week_centered:treatment_conditionShaking treatment -0.014832
## lineageHO_VLN:treatment_conditionShaking treatment 0.089953
## lineageHO_SPT:treatment_conditionShaking treatment 0.239207
## lineageHO_SPC:treatment_conditionShaking treatment 0.349671
## lineageHO_MT:treatment_conditionShaking treatment 0.209042
## lineageHV_GAL:treatment_conditionShaking treatment 0.147370
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.041648
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.049252
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment -0.058406
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.007118
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -0.090908
##                                     Std. Error
## (Intercept)                       0.074015
## week_centered                      0.038461
## lineageHO_VLN                     0.095343
## lineageHO_SPT                      0.088903
## lineageHO_SPC                      0.088717
## lineageHO_MT                      0.086495
## lineageHV_GAL                     0.087060
## treatment_conditionShaking treatment 0.112202
## week_centered:lineageHO_VLN        0.050688
## week_centered:lineageHO_SPT        0.045995
## week_centered:lineageHO_SPC        0.046244
## week_centered:lineageHO_MT        0.044824
## week_centered:lineageHV_GAL        0.045329
## week_centered:treatment_conditionShaking treatment 0.060821
## lineageHO_VLN:treatment_conditionShaking treatment 0.141613
## lineageHO_SPT:treatment_conditionShaking treatment 0.128442
## lineageHO_SPC:treatment_conditionShaking treatment 0.127850
## lineageHO_MT:treatment_conditionShaking treatment 0.125988
## lineageHV_GAL:treatment_conditionShaking treatment 0.128148

```

```

## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.078592
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.070412
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.070488
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.069369
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.070639
## z value
## (Intercept) -1.573
## week_centered 6.968
## lineageHO_VLN -4.615
## lineageHO_SPT 5.418
## lineageHO_SPC 3.580
## lineageHO_MT 6.632
## lineageHV_GAL 5.883
## treatment_conditionShaking treatment -2.429
## week_centered:lineageHO_VLN -0.971
## week_centered:lineageHO_SPT -2.189
## week_centered:lineageHO_SPC -2.736
## week_centered:lineageHO_MT -5.077
## week_centered:lineageHV_GAL -1.930
## week_centered:treatment_conditionShaking treatment -0.244
## lineageHO_VLN:treatment_conditionShaking treatment 0.635
## lineageHO_SPT:treatment_conditionShaking treatment 1.862
## lineageHO_SPC:treatment_conditionShaking treatment 2.735
## lineageHO_MT:treatment_conditionShaking treatment 1.659
## lineageHV_GAL:treatment_conditionShaking treatment 1.150
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.530
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.699
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment -0.829
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.103
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment -1.287
## Pr(>|z|)
## (Intercept) 0.115820
## week_centered 3.21e-12 ***
## lineageHO_VLN 3.94e-06 ***
## lineageHO_SPT 6.04e-08 ***
## lineageHO_SPC 0.000344 ***
## lineageHO_MT 3.30e-11 ***
## lineageHV_GAL 4.04e-09 ***
## treatment_conditionShaking treatment 0.015143 *
## week_centered:lineageHO_VLN 0.331626
## week_centered:lineageHO_SPT 0.028607 *
## week_centered:lineageHO_SPC 0.006213 **
## week_centered:lineageHO_MT 3.84e-07 ***
## week_centered:lineageHV_GAL 0.053602 .
## week_centered:treatment_conditionShaking treatment 0.807336
## lineageHO_VLN:treatment_conditionShaking treatment 0.525295
## lineageHO_SPT:treatment_conditionShaking treatment 0.062549 .
## lineageHO_SPC:treatment_conditionShaking treatment 0.006238 **
## lineageHO_MT:treatment_conditionShaking treatment 0.097072 .
## lineageHV_GAL:treatment_conditionShaking treatment 0.250145
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.596166
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.484246
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.407332
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.918272

```

```
## week_centered:lineageHV_GAL:treatment_conditionShaking treatment 0.198115
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HV_GAL")
summary(glmmTMB(buds ~ week_centered * lineage * treatment_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * treatment_condition + (1 | replicate_unique)
## Data: buds_df
##
##          AIC          BIC    logLik -2*log(L)  df.resid
##   12602.1    12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111  0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept)                Estimate
## week_centered                0.18052
## lineageHC_MT                -0.51214
## lineageHO_VLN               -0.95212
## lineageHO_SPT               -0.03049
## lineageHO_SPC               -0.19453
## lineageHO_MT                0.06152
## treatment_conditionShaking treatment -0.12516
## week_centered:lineageHC_MT    0.08749
## week_centered:lineageHO_VLN   0.03828
## week_centered:lineageHO_SPT  -0.01319
## week_centered:lineageHO_SPC  -0.03905
## week_centered:lineageHO_MT  -0.14007
## week_centered:treatment_conditionShaking treatment -0.10574
## lineageHC_MT:treatment_conditionShaking treatment -0.14737
## lineageHO_VLN:treatment_conditionShaking treatment -0.05742
## lineageHO_SPT:treatment_conditionShaking treatment  0.09184
## lineageHO_SPC:treatment_conditionShaking treatment  0.20230
## lineageHO_MT:treatment_conditionShaking treatment  0.06167
## week_centered:lineageHC_MT:treatment_conditionShaking treatment  0.09091
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment  0.13256
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment  0.14016
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment  0.03250
## week_centered:lineageHO_MT:treatment_conditionShaking treatment  0.09803
##
## Std. Error
## (Intercept)                0.04592
## week_centered                0.02400
## lineageHC_MT                0.08706
```

```

## lineageHO_VLN 0.07561
## lineageHO_SPT 0.06730
## lineageHO_SPC 0.06706
## lineageHO_MT 0.06409
## treatment_conditionShaking treatment 0.06174
## week_centered:lineageHC_MT 0.04533
## week_centered:lineageHO_VLN 0.04081
## week_centered:lineageHO_SPT 0.03482
## week_centered:lineageHO_SPC 0.03514
## week_centered:lineageHO_MT 0.03325
## week_centered:treatment_conditionShaking treatment 0.03593
## lineageHC_MT:treatment_conditionShaking treatment 0.12815
## lineageHO_VLN:treatment_conditionShaking treatment 0.10619
## lineageHO_SPT:treatment_conditionShaking treatment 0.08783
## lineageHO_SPC:treatment_conditionShaking treatment 0.08701
## lineageHO_MT:treatment_conditionShaking treatment 0.08427
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.07064
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.06138
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.05050
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.05060
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.04903
## z value
## (Intercept) 8.618
## week_centered 7.522
## lineageHC_MT -5.883
## lineageHO_VLN -12.593
## lineageHO_SPT -0.453
## lineageHO_SPC -2.901
## lineageHO_MT 0.960
## treatment_conditionShaking treatment -2.027
## week_centered:lineageHC_MT 1.930
## week_centered:lineageHO_VLN 0.938
## week_centered:lineageHO_SPT -0.379
## week_centered:lineageHO_SPC -1.111
## week_centered:lineageHO_MT -4.212
## week_centered:treatment_conditionShaking treatment -2.943
## lineageHC_MT:treatment_conditionShaking treatment -1.150
## lineageHO_VLN:treatment_conditionShaking treatment -0.541
## lineageHO_SPT:treatment_conditionShaking treatment 1.046
## lineageHO_SPC:treatment_conditionShaking treatment 2.325
## lineageHO_MT:treatment_conditionShaking treatment 0.732
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 1.287
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 2.159
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 2.776
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.642
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 1.999
## Pr(>|z|)
## (Intercept) < 2e-16 ***
## week_centered 5.38e-14 ***
## lineageHC_MT 4.04e-09 ***
## lineageHO_VLN < 2e-16 ***
## lineageHO_SPT 0.65048
## lineageHO_SPC 0.00372 **
## lineageHO_MT 0.33714

```

```
## treatment_conditionShaking treatment 0.04265 *
## week_centered:lineageHC_MT 0.05360 .
## week_centered:lineageHO_VLN 0.34833
## week_centered:lineageHO_SPT 0.70479
## week_centered:lineageHO_SPC 0.26649
## week_centered:lineageHO_MT 2.53e-05 ***
## week_centered:treatment_conditionShaking treatment 0.00325 **
## lineageHC_MT:treatment_conditionShaking treatment 0.25015
## lineageHO_VLN:treatment_conditionShaking treatment 0.58872
## lineageHO_SPT:treatment_conditionShaking treatment 0.29573
## lineageHO_SPC:treatment_conditionShaking treatment 0.02007 *
## lineageHO_MT:treatment_conditionShaking treatment 0.46425
## week_centered:lineageHC_MT:treatment_conditionShaking treatment 0.19812
## week_centered:lineageHO_VLN:treatment_conditionShaking treatment 0.03081 *
## week_centered:lineageHO_SPT:treatment_conditionShaking treatment 0.00551 **
## week_centered:lineageHO_SPC:treatment_conditionShaking treatment 0.52069
## week_centered:lineageHO_MT:treatment_conditionShaking treatment 0.04558 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4 Estimated marginal means

4.1 Table

Create a character string containing the 6 exact centered values.

```
week_values <- 1:6 - mean(buds_df$week, na.rm = TRUE)
week_terms <- paste0("week_centered [", paste(week_values, collapse = ","), " ]")
```

Compute the estimated marginal means of weekly bud production for every combination of week, treatment condition, and lineage, based on the selected model.

```
buds_emmeans <-
  as.data.frame(ggemmeans(buds_pois_mod_33,
    terms = c(week_terms, "treatment_condition", "lineage"),
    bias_correction = TRUE)) %>%
  arrange(facet, group, x)
```

Back-transform the centered week to the original scale (weeks 1 to 6) for plotting.

```
buds_emmeans$x <- buds_emmeans$x + mean(buds_df$week, na.rm = TRUE)
```

Round the values to ensure ggplot perfectly recognizes the integers 1 to 6.

```
buds_emmeans$x <- round(buds_emmeans$x)
```

4.2 Plots

4.3 Common functions

Function to plot, for a single lineage, the predicted weekly bud production by treatment condition.

```

buds_plot_treatment <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)

  prefix <- sub("_.*", "", lineage)
  prefix_format <- paste0(substr(prefix, 1, 1), tolower(substr(prefix, 2,
nchar(prefix))))
  label_expr <- parse(text = paste0("bolditalic('", prefix_format, "') * bold('",
sub(".*_", "_", lineage), "')"))

  ggplot(data_sub, aes(x = x, y = predicted, color = group, fill = group)) +
    geom_line(linewidth = 0.8) +
    geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
    labs(x = if(show_x_title) "Age (weeks)" else NULL,
         y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
         else NULL,
         title = NULL) +
    theme_bw() +
    theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
element_blank(),
          axis.ticks.x = if(show_x_text) element_line() else element_blank(),
          axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
element_blank(),
          axis.ticks.y = if(show_y_text) element_line() else element_blank(),
          axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
10)),
          axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
margin = margin(l = 25)),
          axis.line.y.right = element_blank(),
          plot.margin = margin(l = 5, r = 5, t = 5, b = 5),
          legend.position = "none") +
    scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
    scale_y_continuous(position = "right") +
    coord_cartesian(ylim = c(0, 3.5)) +
    scale_color_manual(values = treatment_condition_colors) +
    scale_fill_manual(values = treatment_condition_colors) +
    annotate("text", x = 1.2, y = 3.3, label = label_expr,
            color = "black", size = 5, hjust = 0, vjust = 1) }

```

Function to plot, for a single lineage, the predicted weekly bud production pooled across treatment conditions.

```

buds_plot_lineage <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)
  data_lineage <- aggregate(cbind(predicted, conf.low, conf.high) ~ x, data = data_sub,
FUN = mean)
  data_lineage <- data_lineage[order(data_lineage$x), ]

  prefix <- sub("_.*", "", lineage)
  prefix_format <- paste0(substr(prefix, 1, 1), tolower(substr(prefix, 2,
nchar(prefix))))
  label_expr <- parse(text = paste0("bolditalic('", prefix_format, "') * bold('",
sub(".*_", "_", lineage), "')"))

```

```

ggplot(data_lineage, aes(x = x, y = predicted)) +
  geom_line(linewidth = 0.8, color = "black") +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, fill = "black",
    linewidth = 0.3) +
  labs(x = if(show_x_title) "Age (weeks)" else NULL,
    y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
    else NULL,
    title = NULL) +
  theme_bw() +
  theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
    element_blank(),
    axis.ticks.x = if(show_x_text) element_line() else element_blank(),
    axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
    element_blank(),
    axis.ticks.y = if(show_y_text) element_line() else element_blank(),
    axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
    10)),
    axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
    margin = margin(l = 25)),
    axis.line.y.right = element_blank(),
    plot.margin = margin(l = 5, r = 5, t = 5, b = 5)) +
  scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
  scale_y_continuous(position = "right") +
  coord_cartesian(ylim = c(0, 3.5)) +
  annotate("text", x = 1.2, y = 3.3, label = label_expr,
    color = "black", size = 5, hjust = 0, vjust = 1) }

```

4.4 Panel A: interaction between treatment condition and lineage

```

A1 <- buds_plot_treatment("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
A2 <- buds_plot_treatment("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
A3 <- buds_plot_treatment("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
A4 <- buds_plot_treatment("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
A5 <- buds_plot_treatment("HC_MT", show_y_text = FALSE, show_x_text = TRUE, show_x_title
= TRUE)
A6 <- buds_plot_treatment("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_y_title
= TRUE)

```

Add a letter tag (A) directly to the first sub-plot before combining.

```

A1 <- A1 + labs(tag = "A") + theme(plot.tag = element_text(size = 20, face = "bold"))

```

Combine A1 to A6 into panel A.

```

A <- (A1 + A2 + A3) / (A4 + A5 + A6) + plot_layout(heights = c(1, 1))

```

4.5 Panel B: lineage effect

```

B1 <- buds_plot_lineage("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
B2 <- buds_plot_lineage("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
B3 <- buds_plot_lineage("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
B4 <- buds_plot_lineage("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
B5 <- buds_plot_lineage("HC_MT", show_y_text = FALSE, show_x_text = TRUE, show_x_title =
TRUE)
B6 <- buds_plot_lineage("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_y_title =
TRUE)

```

Add a letter tag (B) directly to the first sub-plot before combining.

```

B1 <- B1 + labs(tag = "B") + theme(plot.tag = element_text(size = 20, face = "bold"))

```

Combine B1 to B6 into panel B.

```

B <- (B1 + B2 + B3) / (B4 + B5 + B6) + plot_layout(heights = c(1, 1))

```

4.6 Assembly

Extract legend from the plot A6 to use later on the full figure.

```

legend_treatment <- cowplot::get_legend(
  A6 +
  labs(color = NULL, fill = NULL) +
  theme(legend.position = "bottom",
        legend.direction = "horizontal",
        legend.text = element_text(size = 13)))

```

Build title graphical objects for each panel.

```

interaction_title_grob <- cowplot::ggdraw() +
  cowplot::draw_label("Interaction between the shaking treatment and the lineage",
                      fontface = "bold", size = 16, x = 0.5, hjust = 0.5)

lineage_title_grob <- cowplot::ggdraw() +
  cowplot::draw_label("Lineage effect",
                      fontface = "bold", size = 16, x = 0.5, hjust = 0.5)

```

Stack the titles, panels, and legend vertically into a single column.

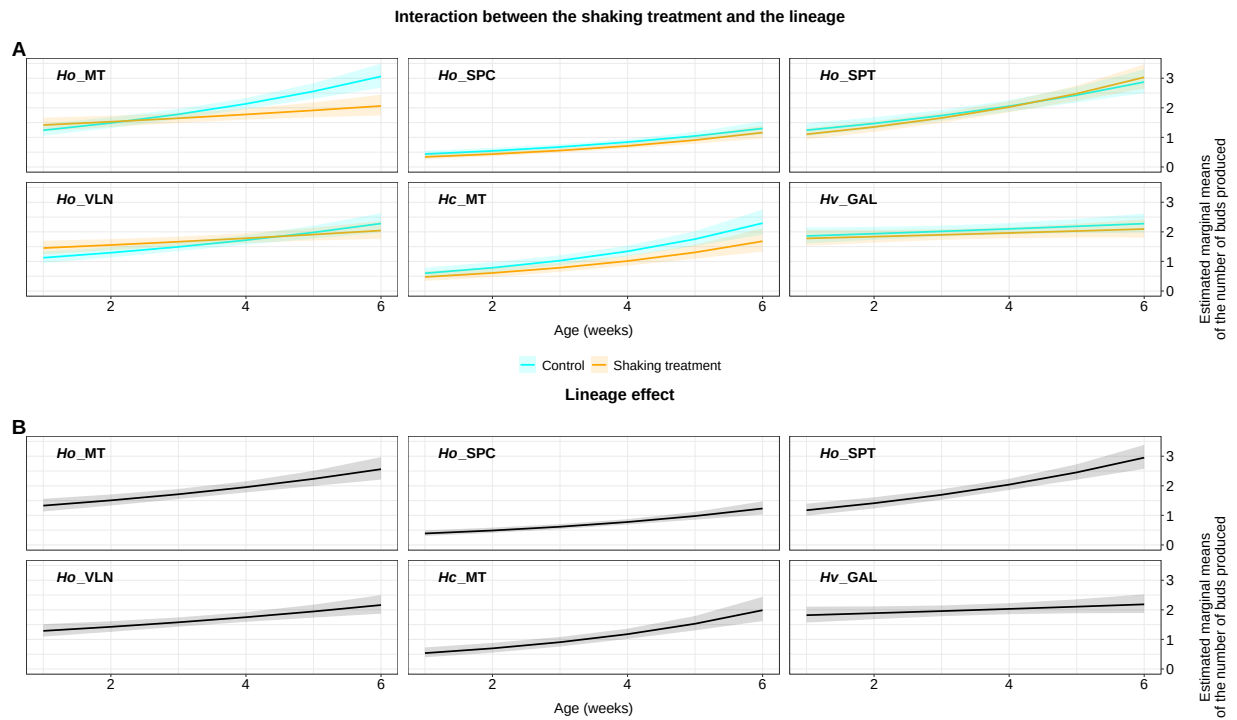
```

final_buds_panel <- cowplot::plot_grid(interaction_title_grob,
                                       A,
                                       legend_treatment,
                                       lineage_title_grob,
                                       B,
                                       ncol = 1,
                                       rel_heights = c(0.08, 1, 0.1, 0.08, 1)) +
  theme(plot.margin = margin(l = 5, r = 5, t = 5, b = 5))

```

Show the final figure.

```
print(final_buds_panel)
```



```
ggsave("figure_5.png", plot = final_buds_panel, width = 21, height = 29.7, units = "cm",
      dpi = 300)
```